Visualizing Epidemic Spread: An Interactive SIR Model for Understanding Disease Transmission

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***Abstract*—This project proposes an interactive scientific visualization that explains how an infectious disease can spread through a population using the Susceptible–Infected–Recovered (SIR) model. The intended audience is college students and members of the general public who have little experience with mathematical epidemiology. The visualization will tell a connected scientific story showing how individuals move between susceptible, infected, and recovered states and how changes in transmission and recovery rates affect a simulated outbreak. The proposed project contains six distinct dynamic views: a population-state view, transmission-mechanism view, animated outbreak view, SIR time-series view, epidemic-peak view, and scenario-comparison view. Users will be able to modify model parameters, inspect values, control simulation time, and compare scenarios. The project will use an explicitly labeled scientific simulation rather than real patient-level data.**

I. Introduction

Infectious disease outbreaks are dynamic processes. The number of infected individuals can increase, reach a peak, and later decrease as infected individuals recover and the susceptible population changes.

The central question of this project is:

> How do transmission and recovery rates influence the size, timing, and shape of an epidemic?

The target audience is undergraduate students and members of the general public without advanced knowledge of epidemiology.

The main objectives are to help users:

1) distinguish susceptible, infected, and recovered population states;
2) understand how transmission and recovery move individuals between states;
3) observe how these populations change over time; and
4) compare epidemic outcomes under different model parameters.

The visualization will follow a beginning, middle, and ending structure. The beginning will introduce the population states and the main scientific question. The middle will show how transmission and recovery cause the outbreak to develop and how those changes appear in quantitative graphs. The ending will allow users to compare scenarios, identify the epidemic peak, and consider the insights and limitations of the model.

II. Related Work

The Susceptible–Infected–Recovered model is based on the mathematical epidemic framework introduced by Kermack and McKendrick. Their work describes how individuals can move between different population states during an epidemic [1].

Educational resources also use interactive models to explain infectious disease transmission. HHMI BioInteractive provides educational materials that allow students to explore how changes in disease characteristics affect the spread of infection [3].

The Centers for Disease Control and Prevention (CDC) also uses epidemic curves to represent the number of illnesses occurring during an outbreak over time [2]. These visualizations help communicate the timing and magnitude of outbreaks.

The proposed project combines these concepts into a connected interactive story. Instead of showing only an epidemic curve, users will see the underlying population states, transmission mechanism, dynamic outbreak, mathematical curves, epidemic peak, and comparisons between scenarios.

III. Methodology and Workflow

A. Scientific Model

The project will use an explicitly labeled SIR scientific simulation.

The population is divided into three compartments:

- $S(t)$ = susceptible individuals,
- $I(t)$ = infected individuals,
- $R(t)$ = recovered individuals.

The total population is

$$N = S(t) + I(t) + R(t). \tag{1}$$

The SIR model is represented by:

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \tag{2}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \tag{3}$$

$$\frac{dR}{dt} = \gamma I. \tag{4}$$

Here, β represents the effective transmission rate and γ represents the recovery rate.

A simplified basic reproduction value can also be represented as

$$R_0 = \frac{\beta}{\gamma}. \tag{5}$$

B. Variables

The primary variables are:

- S: susceptible population, measured in people;
- I: infected population, measured in people;
- R: recovered population, measured in people;
- t: simulation time, measured in days;
- β: effective transmission rate, measured in day^{-1};
- γ: recovery rate, measured in day^{-1}; and
- N: total population, measured in people.

Example starting values may be $N = 1000$, $I(0) = 10$, $R(0) = 0$, and $S(0) = 990$. These numbers are simulation settings rather than observed disease data.

C. Model Assumptions and Limitations

The model assumes a closed population with no births, deaths, or migration during the simulation. It assumes homogeneous mixing and assumes that recovered individuals remain outside the susceptible population during the simulated period.

Real epidemics are more complicated. Disease transmission may depend on age, geography, contact networks, vaccination, behavior, immunity, and other factors. The project will therefore clearly identify the visualization as an educational simulation rather than a prediction of a real epidemic.

An anticipated insight is that users will be able to observe how changes in transmission and recovery rates affect the timing and size of an epidemic peak. However, these are expected educational insights rather than results that have already been obtained.

D. Data and Workflow

The project will generate simulated data with a structure similar to:

```
day, susceptible, infected, recovered
0,   990,         10,       0
1,   ...,         ...,      ...
2,   ...,         ...,      ...
```

The proposed workflow is:

1) Select values for population size, transmission rate, recovery rate, and initial infections.
2) Initialize the susceptible, infected, and recovered populations.
3) Calculate the SIR model across simulation time.
4) Store the population values for each simulated day.
5) Calculate quantities such as peak infections and peak day.
6) Transform the results into the format required by each visualization.
7) Update the connected views whenever the user changes a parameter.

Because the data are generated by the simulation, missing values are not expected. Invalid inputs or calculation errors will be reported rather than silently replaced.

Percentages will be calculated directly from population counts. For example,

$$P_I(t) = \frac{I(t)}{N} \times 100. \tag{6}$$

Percentages from different denominators will not be averaged without justification. Median values are not currently required because the proposed deterministic simulation produces one value for each variable at each time step.

E. Planned Data Preparation and Transformations

JavaScript will be used to prepare the simulation data before it is displayed. The SIR equations will be numerically calculated at fixed time steps. Each generated state will be stored using fields for day, susceptible population, infected population, and recovered population.

The planned data-processing logic will follow a structure similar to:

```
for each simulation time step:
    calculate new S value
    calculate new I value
    calculate new R value
    store day, S, I, and R
```

After the simulation is generated, JavaScript will calculate additional values needed by the visualizations. These transformations will include infected percentage, peak infected population, peak day, final recovered population, and differences between selected scenarios.

The processed values will then be passed to the visualization layer. D3.js may be used to create the charts, update scales, display tooltips, and animate changes when users modify parameters.

SQL is not required for the proposed project because the data will be generated directly by the scientific simulation rather than retrieved from a database.

F. Six-View Storyboard

The project will contain six connected dynamic views. Each view has a different explanatory purpose.

1) View 1: Population State: **Question:** What portion of the population is susceptible, infected, or recovered?

Visual: Population icons or circles representing people. Each state will have a different visual encoding and label.

Why it fits: A population-based visual makes the three SIR categories easier to understand before the user encounters equations and graphs. It gives an intuitive representation of how many people belong to each state.

Variables: S, I, R, and N.

Interaction: Users can hover over a population state to see its exact count and percentage. A time slider will change the displayed population state.

Resulting change: Moving the time slider will update the displayed numbers and icons so the user can see how the population composition changes during the outbreak.

Connection: This introduces the three states used throughout the remaining views.

VIEW 1: POPULATION STATE

S S S I I R

Susceptible – Infected – Recovered

Time Slider: [————-]

2) View 2: Transmission Mechanism: **Question:** How does a person move between the SIR states?

Visual: A flow diagram representing:

$$\text{Susceptible} \rightarrow \text{Infected} \rightarrow \text{Recovered}$$

Why it fits: A flow diagram clearly communicates the direction of movement between compartments and connects the mathematical parameters to the disease-transmission mechanism.

Variables: S, I, β, and γ.

Interaction: Users can change β and γ with sliders.

Resulting change: Changing β or γ will recalculate the simulation. Increasing or decreasing these parameters will change the rate at which the simulated population moves between states and update the connected visualizations.

Connection: The mechanism shown here produces the changing population shown in View 3.

VIEW 2: TRANSMISSION

Susceptible $\longrightarrow$ Infected $\longrightarrow$ Recovered

β γ

Transmission: [———-]

Recovery: [————-]

3) View 3: Animated Outbreak: **Question:** What does the simulated outbreak look like as time progresses?

Visual: An animated population of dots or icons whose states change as the simulation advances.

Why it fits: Animation makes the time-dependent nature of an epidemic visible and allows users to observe how the population changes instead of only seeing the final outcome.

Variables: t, S, I, and R.

Interaction: Users can play, pause, step forward, or reset the simulation.

Resulting change: Playing or stepping the simulation advances time and updates the displayed SIR states. Pausing freezes the current state, while reset returns the simulation to its initial values.

Connection: This view provides an intuitive representation of the numerical values shown in View 4.

VIEW 3: OUTBREAK ANIMATION

S S I S R

S I I S R

[Play] [Pause] [Step] [Reset]

4) View 4: SIR Curves: **Question:** How do the three populations change over time?

Visual: A line chart containing susceptible, infected, and recovered curves.

Why it fits: A line chart is appropriate for showing continuous change over time and allows users to compare the behavior of the three populations on the same time scale.

Variables: $S(t)$, $I(t)$, $R(t)$, and time in days.

Interaction: Users can hover over a day to inspect exact population values. A time cursor will synchronize this chart with the population animation.

Resulting change: Moving the time cursor will highlight the selected day and update the corresponding population state in the other connected views.

Connection: The infected curve will be examined more closely in View 5.

VIEW 4: SIR CURVES

Population

|\ / R

| \ /\ I

| \/

|_____ Time

Hover to inspect values

5) View 5: Epidemic Peak: **Question:** When does the number of infected people reach its maximum?

Visual: An epidemic curve highlighting the maximum infected population and the corresponding simulation day.

Why it fits: An epidemic curve makes the rise, peak, and decline of infections easy to identify and directly supports comparison of epidemic magnitude and timing.

Variables: $I(t)$, peak infections, and peak day.

Interaction: Users can hover over the curve and modify β or γ.

Resulting change: Hovering will display the exact infected population for a selected day. Changing β or γ will recalculate the curve and may change both the size and timing of the peak.

Connection: After examining one epidemic, the user proceeds to View 6 to compare multiple scenarios.

VIEW 5: EPIDEMIC PEAK

Infected

| /\ ← Peak

| / \

|_/ ____

|_____ Day

6) View 6: Scenario Comparison: **Question:** How do different transmission and recovery rates change the epidemic?

Visual: Two epidemic curves or panels representing different parameter combinations.

Why it fits: A side-by-side comparison allows users to directly evaluate how changing model parameters affects peak size, peak timing, and the final population outcome.

Variables: β, γ, R_0, peak infections, peak day, and final recovered population.

Interaction: Users can select two scenarios and compare their curves and summary values. A reset button will restore the baseline settings.

Resulting change: Selecting different parameter combinations will update both scenario curves and their summary statistics, allowing users to directly compare the outcomes.

Connection: This final view combines the concepts introduced throughout the previous five views and completes the scientific story.

VIEW 6: SCENARIO COMPARISON

Scenario A	Scenario B
$\beta =$ value	$\beta =$ value
$\gamma =$ value	$\gamma =$ value

Peak A vs. Peak B
Peak Day A vs. Peak Day B

[Reset]

G. Interaction and Visual Design

The project will include several meaningful interactions:

1) sliders for changing β and γ;
2) play, pause, step, and reset controls;
3) a time slider;
4) hover tooltips displaying actual values; and
5) controls for comparing two simulation scenarios.

Labels and units will be displayed consistently. Population counts will be measured in people and time in days. Visual encodings will remain consistent between views. Color will not be the only way to distinguish population states; text labels, symbols, or line styles will also be used.

All quantitative graph axes will be clearly labeled with their units. Linear scales will be used for population and time unless another scale is scientifically justified. Axes will begin at zero when appropriate, and scenario comparisons will use consistent scales so that visual differences are not exaggerated. Tooltips will display actual numerical values rather than requiring users to estimate values from the graphs.

H. Implementation Plan

The proposed implementation will use HTML, CSS, and JavaScript. D3.js may be used to create the interactive charts and animations.

The SIR equations will be numerically evaluated across time. The resulting values will then be shared between the six visualizations.

A common simulation-time variable will synchronize the views. For example, selecting day 20 in the SIR graph will cause the population visualization to display the corresponding state for day 20.

The implementation will separate simulation calculations from visualization code. This will make it easier to test the mathematical model independently before connecting it to the interactive views.

IV. Timeline and Evaluation

The project will be completed in several stages:

1) Finalize the model, assumptions, and visual designs.
2) Implement and test the SIR simulation.
3) Develop Views 1–3.
4) Develop Views 4–6.
5) Connect the interactions between views.
6) Test, evaluate, and revise the final visualization.

A. Numerical Check

A numerical validation will verify population conservation. At each simulation step,

$$S(t) + I(t) + R(t) = N. \tag{7}$$

The program will calculate

$$\epsilon(t) = |N - [S(t) + I(t) + R(t)]|. \tag{8}$$

This value should remain approximately zero, within numerical tolerance.

The simulation will also verify that the susceptible population does not increase and that the recovered population does not decrease under the standard model assumptions.

B. Boundary-Case Check

One boundary test will use

$$\beta = 0. \tag{9}$$

When transmission is zero, susceptible individuals should not become infected. The initially infected population should only move into the recovered population.

A second boundary case will use

$$I(0) = 0. \tag{10}$$

Without an initially infected individual, the model should not generate an epidemic.

C. Evaluation and Revisions

Users will be asked to interact with the visualization and answer questions such as:

- Which parameter affects how quickly infections increase?
- On what day does the infected population reach its peak?
- How does changing the recovery rate affect the epidemic curve?
- How does the animated population correspond to the SIR graph?

Feedback about confusing labels, controls, transitions, or visualizations will be recorded. Repeated problems identified

during testing will be used to revise the interface. For example, unclear controls may be relabeled, confusing visual encodings may be changed, and instructions may be added when users have difficulty understanding an interaction.

V. Contingency and Collaboration

One technical risk is that animating a large number of individual population icons could cause performance problems. If this occurs, the project will display a smaller representative population while maintaining the correct aggregate SIR values.

If synchronizing continuous animations becomes too complicated, a shared time slider will be used instead. This fallback will still preserve all six distinct explanatory views.

Another possible challenge is that users may interpret the simulation as a prediction of a real disease. To reduce this risk, the project will clearly label the visualization as a simplified scientific simulation and display the major model assumptions and limitations.

This project will be completed individually. The work will be organized into model development, data preparation, visualization development, interaction design, testing, documentation, and final integration. This organization will make it easier to complete and evaluate each part separately before combining the six views into the final project.

VI. Conclusion

This project proposes a six-view interactive scientific story explaining epidemic spread using the SIR model. The visualization will begin by introducing susceptible, infected, and recovered population states. It will then explain transmission, animate the simulated outbreak, display quantitative SIR curves, identify the epidemic peak, and allow users to compare different scenarios.

The project is intended to help users connect individual disease transmission with population-level epidemic behavior. Interactive parameter controls, time controls, tooltips, and scenario comparisons will allow users to actively explore the model.

The anticipated insight is that users will better understand how changes in transmission and recovery rates influence the simulated epidemic. However, the project will also emphasize the limitations of the simplified SIR model and will not present the simulation as a prediction of a specific real-world epidemic.

AI Assistance Disclosure

Generative artificial intelligence (ChatGPT) was used to assist with brainstorming the project topic, organizing the proposal, explaining the SIR model, and drafting portions of the written proposal and storyboard plan.

AI-generated material was reviewed against the assignment requirements. The SIR equations and scientific concepts were checked against the cited scientific and educational sources, and the references were reviewed for relevance to the proposed visualization. The final proposal was also compared with the assignment instructions to verify that the required sections, six-view storyboard, interactions, model variables, evaluation plan, contingency plan, and AI disclosure were included. The author remains responsible for the final content, citations, implementation decisions, and accuracy of the project.

References

[1] W. O. Kermack and A. G. McKendrick, "A contribution to the mathematical theory of epidemics," *Proceedings of the Royal Society of London. Series A*, vol. 115, no. 772, pp. 700–721, 1927.

[2] Centers for Disease Control and Prevention, "How to Read an Epi Curve." [Online]. Available: https://www.cdc.gov/foodborne-outbreaks/outbreak-basics/how-to-read-an-epi-curve.html

[3] Howard Hughes Medical Institute BioInteractive, "Modeling Disease Spread." [Online]. Available: https://www.biointeractive.org/classroom-resources/modeling-disease-spread